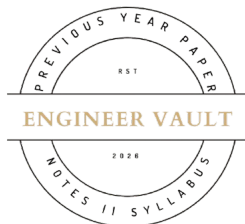


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SUBJECT CODE NO: - RNP-8009
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group A & B) CSE / AI & ML / EEE (NEP) Sem - I
Examination June/July 2025
Basics of Electrical Engineering

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Question No. 2, 3, 4, 5, 6 and 7.

SECTION A**20 Marks****Q1 Solve any 10 from the following questions.**

1. The algebraic sum of voltages around any closed path in a network is equal to

- A. ZERO B. Infinite C. low D. high

2. The resistance of a conductor is proportional to its _____

- A. Length B. Area C. current D. power

3. Line voltage in star connected three phase _____ times phase voltage.

- A. 3 time B. $\sqrt{3}$ C. Equal D. 2 Times

4. Line current in delta connected three phase _____ times line current.

- A. 3 time B. $\sqrt{3}$ C. Equal D. 2 Times

5. The value of form factor of sinusoidal waveform is _____

- A. 1.11 B. 1.21 C. 0.70 D. 0.63

6. RMS value of sinusoidal wave form is _____ Maximum value.

- A. 1.11 B. 1.21 C. 0.70 D. 0.63

7. The conductor and flux having angle between them 90 then flux link

- A. Maximum B. Minimum C. Equal D. Zero

8. The ratio of maximum value to the r.m.s. value of an alternating quantity is known _____

- A. Zero B. Peak Factor C. Form Factor D. None

9. The flux variations being brought about by variations

- A. Density B. voltage C. current D. Power

10. An ideal voltage source is that which has an internal resistance

- A. Zero B. Low C. High D. Infinite

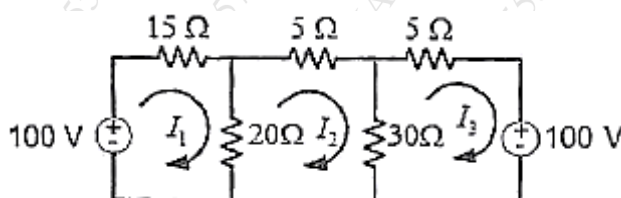
11. The phenomenon of lagging of flux density (B) behind the magnetising force (H) in a magnetic material
 A. Flux B. Current C. B-H Curve D. Voltage
12. In Kirchhoff's first law $\sum i = 0$ at the junction is based on the conservation of
 A. Charge B. Current C. Energy D. Density
13. The resistance of a material is most commonly determined by four factors-length, cross-sectional area, type of material and _____
 A. Current B. Voltage C. Power D. Temperature

SECTION B**40 Marks**

Q2 a) Prove the Series and parallel combination of resistor with circuit diagram. **05**

b) Expression for value of α at different temperature in detail. **05**

Q3 a) Find loop Current I_1 , I_2 & I_3 By Mesh Analysis. **05**



b) Explain Superposition theorem with an example. **05**

Q4 a) Comparison between Magnetic and Electric Circuits. **05**

b) A solenoid have 800 turns supply with 30 volt of resistance is 6 ohm, length of solenoid is 16 cm with area is 6 sq.cm, relative permeability is 800 find
 1) MMF 2) Magnetic field intensity 3) Flux density 4) Reluctance **05**

Q5 a) Prove the Coefficient of Coupling with equation. **05**

b) Prove the Capacitors In Series with detail. **05**

Q6 a) Explain R-L Series A.C. Circuit with phasor diagram **05**

b) Explain R-C Series A.C. Circuit. **05**

Q7 a) Explain the relation between Star connected phase voltage and line voltage in detail **05**

b) Explain the relation between Delta connected phase current and line current in detail **05**